Contact Details

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Employment & Education

Ph.D. in Mathematics, current position since 2022
Northwestern University

B.S. in Mathematics, 2018 – 2022
Massachusetts Institute of Technology

Research Papers

$\mathbb{C}$-*Motivic Eilenberg-Mac Lane Spectra as* MGL-*Motivic Thom Spectra and Normed* $\mathbb{G}_m$-*Hochschild Homology*
preprint, arXiv:2610.26771

Generalized Thom Spectra, Conjugate Stacks, and truncated Brown-Peterson Spectra
preprint, arXiv:2610.18754

Finding Bounded Simplicial Sets with Finite Homology with Peter Rowley
preprint, arXiv:2112.12230

Projective Toric Varieties of Codimension 2 with maximal Castelnuovo–Mumford regularity with Alan Peng, and Vijay Srinivasan
Journal of Pure and Applied Algebra, vol. 227, is. 1, Jan. 2023, 107162

Biconed Graphs, Weighted Forests, and H-Vectors of Matroid Complexes with Anton Dochtermann, Evan Haithcock, Joshua Marsh, Suho Oh, and Anna Truman
The Electronic Journal of Combinatorics, vol. 28, no. 4, 19 Nov. 2021, 10.37236/9849.

Honors

Simons Dissertation Graduate Research Fellowship, 2025 – 2027

NSF Graduate Research Fellowship, 2022 – 2027

Invited Talks

AMS Special Session on Recent Developments in Homotopy Theory at the 2027 Joint Mathematics Meetings, 2027 January (to occur)

University of Chicago Topology Semianr, 2026 October (to occur)

Univerisity of Washington Topology Seminar, 2026 March

University of Rochester Topology Seminar, 2026 January

Texas State REU Visitor Lecture, 2020 July

Contributed Talks	*Böksdedt periodicity and the even filtration*, Conference on "New Perspectives on Stable Homotopy and Beyond", Bonn, 2026 September
	Short Communication, ICM 2026, Philadelphia, 2026 July
	$\mathbb{E}_3$-MU-*Thom Spectra*, Young Topologists Meeting 2026, Copenhagen, 2026 July
	K(1)-local Power Operations, European Talbot 2025, Kolding, 2025 July
	Syntomic Cohomology of $\mathbb{Z}$ and ℓ, Talbot 2024, Nacadoches, 2024 August
	Finding Bounded Simplicial Sets with Finite Homology, SPUR Conference at Massachusetts Institute of Technology, online , 2021 August
	Projective Toric Varieties of Codimension 2 with maximal Castelnuovo–Mumford regularity, Joint Mathematics Meeting 2021 Undergraduate Poster Session, online, 2021 January
	Equality in the Eisenbud–Goto Conjecture for Certain Toric Varieties, Undergraduate Mathematics Symposium at University of Illinois at Chicago, online, 2020 November
	Bi-coned graphs, edge rooted forests, and Stanley's h-vector conjecture, Joint Mathematics Meeting 2021 Undergraduate Poster Session, Denver, 2021 January
Organization	Synthetic Spectra Reading Seminar, Northwestern University, 2024 Spring
Expositional Seminar Talks	Prismatic Cohomology of Ring Spectra, University of Washington, 2026 March
	τ-invertible Synmthetic Spectra, Synthetic Spectra Reading Seminar, Northwestern University, 2024 May
	Overview, Synthetic Spectra Reading Seminar, Northwestern University, 2024 March
	Section 3, Seminar on the Chromatic Nullstellensatz, Northwestern University, 2023 April
Teaching	TA, 18.03: Differential Equations, Massachusetts Institute of Technology, 2021 Fall
	TA, 18.02: Multivariable Calculus, Massachusetts Institute of Technology, 2021 Spring
	TA, 18.s097: Introduction to Proof Writing, Massachusetts Institute of Technology, 2021 January
	TA, 18.03: Differential Equations, Massachusetts Institute of Technology, 2020 Fall
Outreach	Math Circle volunteer, Evanston Math Circle, Northwestern University

Mentor, Directed Reading Program, Northwestern University, 2024 Fall

Mentor, Directed Reading Program, Northwestern University, 2024 Winter and Spring

Mentor, Directed Reading Program, Northwestern University, 2023 Fall

Knot Theory Group mentor, PRIMES Circle, Massachusetts Institute of Technology, 2022 Spring

Knot Theory Group mentor, PRIMES Circle, Massachusetts Institute of Technology, 2021 Spring